

Part II: Open problems

1. (20 points) Suppose that AMD has a monopoly in the market for computer chips. AMD's total costs are given by $TC(X) = \frac{5}{2}X^2 + 15,000$. Thus, AMD's marginal cost is $MC(X) = 5X$.
- (a) (10 points) The demand for computer chips is $X_D = 100 - 0.1P$. Find the level of output that maximizes AMD's profits. What price is AMD charging? What are its profits?

Rubric:

- Getting the demand: $P = 1,000 - 10X$ 1 point
- Getting MR: $MR = 1,000 - 20X$ 1 point
- Just Equating MR to MC 1 point
- Getting the final monopoly quantity of 40, 2 points.
- Computing the monopoly price using the demand 3 points.
- Computing profits 2 points, if they used an incorrect quantity or price to compute the profits but the calculation is ok then 1 point.

AMD is producing **40 (5 points)** computer chips, charging a price of **600 (3 points)** per chip and its profits are **5,000 (2 points)**.

- (b) (10 points) What level of output would maximize total surplus in the computer chip market? Calculate the dead-weight loss generated by AMD behaving as a monopolist.

Rubric:

- Setting Demand equal to MC to get efficient quantity 2 points.
- Solving for the efficient quantity 3 points.
- Obtaining the remaining number required to calculate the dead-weight loss by computing marginal cost at the monopoly quantity: $MC(40) = 200$ 2 points.
- Calculating the loss, 3 points.

The efficient level of output is **66.67 or $\frac{200}{3}$ or $\frac{1000}{15}$ (5 points)** and the dead-weight loss is **5334 (5 points)**.

2. (20 points) The demand for flu shots this season is given by $P = 13 - 0.0005Q$. The marginal cost of a flu shot is given by $MC = 8 + 0.0005Q$.

(a) (5 points) In a competitive market, what are the equilibrium price and quantity of flu shots?

Rubric:

- Equating MB to MC to get the market outcome 1 point.
- Solving for the market quantity 2 points.
- Obtaining the market price 2 points.

The competitive equilibrium price is 10.5 (2 points) and the equilibrium quantity is 5000 (3 points).

- (b) (10 points) Suppose that the *marginal external benefit* of flu shot is $MEB = 0.00025Q$. Compute the socially efficient quantity of flu shots.

Rubric:

- Obtaining MSB by adding up MB and the marginal external benefit, 5 points.
 - If they just say that $MSB = MB + MEB$ but mess up the calculation, 3 points.
- Obtaining the efficient quantity by equating MSB to MC 5 points.
 - If they just equate MSB to MC but mess up calculation, 3 points.

The socially efficient quantity of flu shots is $6,666.67$ (10 points).

- (c) (5 points) Suppose that the government decides to give a subsidy of S per flu shot. What subsidy S should the government set to achieve the social optimum?

Rubric:

- There are several ways to get the subsidy which are correct: Evaluating the marginal external benefit at the efficient quantity $MEB(6666.67) + 1.67$ is one.
- Another one is to obtain the difference between MSB and MB at the efficient quantity $MBS(6666.67) = 11.33$ and $MB(6666.67) = 9.66$ whose difference is 1.67.
- Both methods are correct.
- Any variation of these methods that yields the correct tax is also ok.
- If they do something similar to the above, but mess up the calculation, then only 3 points.
- No points if nothing resembles any of the above.

The per-flu shot subsidy that the government needs to put is 1.67 (5 points).